CyberSploit 2

Professional Penetration Testing Project Report & Walkthrough

TARGET	192.168.29.79
OPEN SERVICES	22/tcp SSH • 80/tcp HTTP
INITIAL ACCESS	shailendra via SSH
ESCALATION	Docker → host filesystem → root
FINAL RESULT	Full root-level compromise

Controlled vulnerable-lab assessment documented from the commands, outputs, and web-source evidence supplied during the exercise.

1. Executive Summary

CyberSploit 2 was assessed from initial reconnaissance through root-level compromise. Nmap identified SSH and HTTP. Web-source inspection exposed credential-related information and a ROT47 clue, leading to valid SSH access as shailendra. A local hint identified Docker. Docker was then used to expose the host filesystem to a container and obtain a root shell through chroot. The final proof of compromise was /root/flag.txt.

Assessment outcome: FULL ROOT-LEVEL COMPROMISE.

2. Scope & Methodology

Target: 192.168.29.79 Environment: Controlled vulnerable lab
Method: Reconnaissance → Service Enumeration → Web Enumeration → Credential Discovery → Initial Access → Local Enumeration → Privilege Escalation → Validation.

3. Key Findings

ID	Finding	Impact
F-01	Credential-related information exposed in web content	Initial access
F-02	ROT47 used as simple obfuscation	Credential disclosure
F-03	SSH exposed with usable credentials	Remote access
F-04	Docker capability enabled host filesystem access	Critical / root compromise

4. Reconnaissance

Command: nmap -sV 192.168.29.79
Result: 22/tcp SSH OpenSSH 8.0; 80/tcp HTTP Apache 2.4.37 (CentOS). HTTP was prioritized for information gathering.

5. Web Enumeration & ROT47

The CyberSploit2 page contained a Username / Password / Handle table. The supplied HTML confirms this structure. □filecite□turn1file0□L60-L67□ One row contained encoded-looking values, and the source explicitly included a ROT47 marker. □filecite□turn1file0□L89-L93□ □filecite□turn1file0□L130-L135□

Encoded-looking row: D92:=6?5C2 / 4J36CDA=@:E`. The lab clue directed the credential-analysis process toward ROT47.

6. Initial Access

The discovered username shailendra was used against the exposed SSH service:

```
ssh shailendra@192.168.29.79
```

Authentication succeeded and returned an interactive shell: [shailendra@localhost ~]$. This established the initial foothold.

7. Local Enumeration

The home directory contained hint.txt. Reading it returned docker, identifying Docker as the next investigation area.

8. Docker-Based Privilege Escalation

The demonstrated lab command was:

```
docker run -v /:/mnt --rm -it alpine chroot /mnt bash
```

The -v /:/mnt option mounted the host root filesystem at /mnt inside the container. The chroot /mnt bash portion changed the shell root to that mounted filesystem. The resulting shell was verified with whoami → root.

9. Root Validation & Flag

The /root directory contained flag.txt. Reading the file produced the lab proof-of-compromise banner:

```
Pwned CyberSploit2 POC
```

This confirmed successful completion of the intended privilege-escalation path.

10. Security Impact & Recommendations

The demonstrated chain resulted in complete root-level control. An attacker following the same path could potentially access protected data, modify system configuration, and establish persistence.

Recommendations: remove credentials from public web content; never use ROT47 as password protection; restrict Docker administrative privileges; harden SSH access; monitor container creation and sensitive host mounts; keep the OS and exposed services patched.

11. Final Attack Chain

A compact visual sequence of the complete lab compromise is provided below.

01 RECON	02 NMAP	03 WEB
192.168.29.79	22 SSH • 80 HTTP	Source / credential table
04 ROT47	**05 SSH**	**06 HINT**
Encoding clue	shailendra	hint.txt → docker
07 DOCKER	**08 ROOT**	**09 FLAG**
Host / → /mnt	chroot → whoami=root	/root/flag.txt

Final Status

CYBERSPLOIT 2 SUCCESSFULLY COMPROMISED
Attack path: Nmap → Web Enumeration → ROT47 → SSH → shailendra → Docker → chroot → root → flag.txt.

End of Report